

JEE MAIN REVISION MASTER: PROBABILITY SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 232 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Probability Distribution & Binomial Distribution	120 PYQs	Random Variables
Very High	Conditional Probability & Bayes' Theorem	113 PYQs	Conditional & Bayes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Probability Distribution & Binomial Distribution	$\text{Mean } \mu = \sum x_i p_i, \quad \text{Var } \sigma^2 = \sum x_i^2 p_i - \mu^2, \quad \text{Binomial: } P(X = r) = {}^n C_r p^r q^{n-r}, \quad \text{Mean} = np, \quad \text{Var} = npq$
Conditional Probability & Bayes' Theorem	$P(A B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A) \cdot P(B) \text{ (independent)}, \quad P(E_i A) = \frac{P(E_i)P(A E_i)}{\sum P(E_j)P(A E_j)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Probability Distribution & Binomial Distribution (Very High Yield — 120 PYQs)

Core Principle: Discrete probability distributions, expected value $E(X)$, variance, and Binomial Bernoulli trials ($q = 1 - p$).

■ **Caution / JEE Pitfall:** In a Binomial distribution, the variance (npq) is ALWAYS strictly less than the mean (np) because $q < 1$.

■ **Speed Trick:** Variance formula shortcut: $\text{Var}(X) = E(X^2) - [E(X)]^2$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the ent..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. ..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of get..." [Ans: N/A]

3.2 Conditional Probability & Bayes' Theorem (Very High Yield — 113 PYQs)

Core Principle: Multiplication rule of probability, test for statistical independence, law of total probability, and inverse probability via Bayes' theorem.

■ **Caution / JEE Pitfall:** Mutually exclusive events ($A \cap B = \emptyset$ implies $P(A \cap B) = 0$) CANNOT be independent if both have non-zero probabilities!

■ **Speed Trick:** For independent events: $P(A \cup B) = 1 - P(A')P(B')$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each draw..." [Ans: N/A]
- **[2026 Evening Shift] Q#2 (MCQ):** "A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that..." [Ans: N/A]
- **[2026 Evening Shift] Q#2 (MCQ):** "Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is:

- (A) $(7 / 10)$
- (B) $(10 / 17)$ [Correct]
- (C) $(12 / 19)$
- (D) $(7 / 19)$

Correct Answer: B

Solution Analysis: Let A = event that the letter came from ANANTPUR K = event that the letter came from KANPUR B = event that on the envelope, the two visible consecutive letters are AN We have to find: $P(A|B)$ Using Bayes' theorem, $P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(K)P(B|K)}$ Since the letter is known to have come either from ANANTPUR or KANPUR, and no other information is given, we take $P(A) = P(K) = (1 / 2)$ So, $P(A|B) = \frac{P(B|A)}{P(B|A) + P(B|K)}$ Now we find these probabilities. 1. Probability of seeing AN in ANANTPUR Word: ANANTPUR Its consecutive pairs are: AN, NA, AN, NT, TP, PU, UR There are 7

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Morning Shift (Q#2)

From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $a - b \leq 10$ is $(m / n), \gcd(m, n) = 1$, then $m + n$ is equal to ____.

Correct Answer: 311

Solution Analysis: Let $b = 1 \Rightarrow a > 11 \Rightarrow a$ can take 90 values from 11, 12, ..., 100 Let $b = 2 \Rightarrow a > 12 \Rightarrow 89$ values and so on till $b = 99 \Rightarrow a$ can only take $a = 100$, only 1 value $\Rightarrow$ favourable cases $= 1 + 2 + 3 + \dots + 90 = \frac{(90 \times 91)}{2} = 4095$ Total cases $= 100 \times 99 = 9900$ $\Rightarrow m + n = 311$